8.5 Answers: estimating and the calculator

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For the parent: in Set B, any sensible estimate made with friendly numbers is correct. The answers below show one good estimate and the exact answer, so you can see how close it was.

Set A: Rounding

1. **50.** 47 is nearer to 50 than to 40.
2. **400.** Tens digit 8 → round up.
3. **2 000.** Hundreds digit 6 → round up.
4. **4.** 3,7 is nearer to 4.
5. **12.** The tenths digit is 4 → round down. (Don't round twice: 12,49 → 12,5 → 13 is a common slip.)
6. **€5.** €4,95 is 5 cents from €5.
7. **2,3.** The hundredths digit is 4 → round down.
8. **7,9.** Exactly halfway between 7,8 and 7,9 → round up.
9. **16 000.** Exactly halfway → round up.
10. **0,67.** Thousandths digit 6 → round up.
11. **60.** 63 is nearer to 60.

Set B: Estimate

	Friendly sum	Estimate	Exact (for comparison)
1	50 + 30	80	80
2	200 + 400	600	601
3	6 × 50	300	294
4	20 × 20	400	399
5	800 : 4	200	203
6	€3 + €4 + €2	€9	€9,03
7	4 000 : 20	200	199
8	50 × 10 (or 52 × 10 + 52)	500 (or 572)	572
9	5 × €3 = €15	**Yes**, €20 is enough	€14,95
10	half of 300	150	147

Set C: Does it make sense?

1. **Right.** 40 × 20 = 800, close to 798.
2. **Wrong.** About 450 + 400 = 850, not 7 000. Correct: 845.
3. **Right.** 1 200 : 4 = 300.
4. **Wrong.** About 3 × €20 = €60. Correct: €59,85 (the comma was in the wrong place).
5. **Wrong.** 50 × 5 = 250, not 2 500. Correct: 50.
6. **Right.** 70 × 50 = 3 500, close to 3 456.
7. **Right.** 9,9 + 0,1 = 10.
8. **Wrong.** 15% is less than the whole €80, so it can't be €120. Correct: €12.
9. **Wrong.** 2 cm is less than a finger is wide. A door is about 2 **m** high.
10. **Right.** 28 : 4 = 7.

Set D: Head, paper or calculator?

The letters are suggestions; any choice with a sensible reason is fine.

1. **H: 99.**
2. **H: 100.** Four quarters make a hundred.
3. **P or C: 304.** Estimate: 3 600 : 12 = 300.
4. **P: 722.**
5. **H: 12.** 8 + 4 (half of 8).
6. **C: €417,41.** Estimate: about 18% of €2 300 ≈ 20% of €2 000 = €400. The calculator shows 417.41.
7. **H: 1 998.** 1 000 + 1 000 − 2.
8. **C or P: 8 760 hours.** 365 × 24. Estimate: 400 × 25 = 10 000, a bit high because both numbers were rounded up.
9. **H: 423.** Half of 846.
10. **C: €87,43.** Estimate: 7 × €12,50 = €87,50.

Set E: Reading the calculator

1. **32 minibuses.** 31 are full and the last 2 children still need a seat, so round **up**.
2. **€11,90 each.** 11.9 on the screen means €11,90.
3. **0.3333333 → 0,33.**
4. **About €6,67 each.** 20 : 3 = 6,666… → €6,67. But 3 × €6,67 = €20,01, one cent too much, so one friend pays €6,66. Money doesn't always share out exactly.
5. **A basic calculator may show 14; the correct answer is 11.** Multiply first: 4 × 2 = 8, then 3 + 8 = 11. Many phone and all scientific calculators show 11.
6. **€9,95.** 2,5 × €3,98 = €9,95. Estimate: 2,5 × €4 = €10.
7. **52 full weeks.** The calculator shows 52.142857…; the part after the point is less than a full week (1 day).
8. **13 boxes needed; 12 completely full.** 100 : 8 = 12.5; 12 full boxes hold 96 eggs, the 13th box holds the last 4.
9. **5.** Nice and simple.

Puzzle corner

1. **Less, by 199.** 100 × 99 = 9 900; take away one more 99 → 9 801. 10 000 − 9 801 = 199.
2. **It depends on the pulse.** Example: 20 beats in 15 seconds → 80 per minute → 80 × 60 × 24 = 115 200 beats a day. Anything from about 100 000 to 140 000 is a sensible answer for a resting child.
3. **111 111 111, 222 222 222, 333 333 333.** 12 345 679 × 9 = 111 111 111, so multiplying by 18 (2 × 9) gives 2 × that, and so on. (Notice the 8 is missing from 12345679.)